UTM UNIVERSITI TEKNOLOGI MALAYSIA | Faculty of Computing

Course Code	:	**SECR2213**
Course Name	:	**NETWORK COMMUNICATION**
Lecturer	:	Dr. Adnan Khalid
Academic Session	:	**SEM2_2526**
Assessment Title	:	Network Design for Faculty of Computing Block N28B
Submission Due Date	:	2 April 2026, Thursday

Prepared by :

Matric No.	Student Name
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Assessment Mode : Group (The Chill Trio)

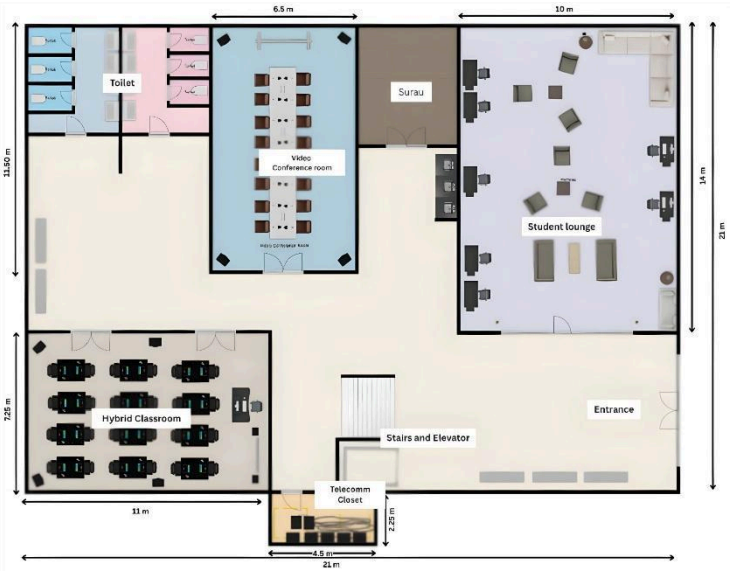
Feedback from Lecturer :

MAIN REPORT

Suggested Floor Plan:

Floor Plan Description: This floor plan is designed to meet the Faculty of Computing's requirements to accommodate the anticipated 15% growth in students and academic staff. The new 2-storey building utilizes a modern **cantilevered architectural design**. The first floor maintains a compact footprint of 21m x 21m to optimize the ground-level area, while the second floor expands to a larger 29.3m x 28m footprint. This overhang provides natural shade to the ground floor entrance and surrounding pathways.
To optimize traffic and functionality, the floors are segmented as follows:

First Floor (21m x 21m): Dedicated to collaborative and communal spaces. The ground floor layout is designed for high accessibility and accommodates high-traffic and flexible learning areas. It includes the required Student Lounge (14 m x 10m), the Hybrid Classroom, and the Video Conferencing Room. Crucially, the Main Distribution Facility (MDF) / Telecom Room is strategically located near the stairs and entrance for optimal backbone connectivity to the 2nd floor and the main campus network.



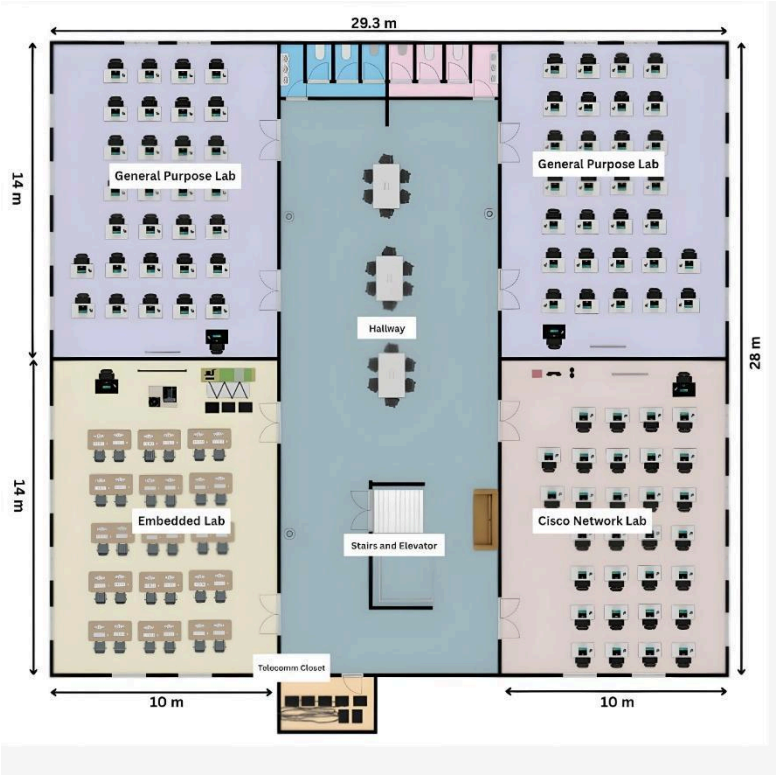
- **Student Lounge:** Placed on the right wing, measuring exactly the required 14m x 10m. It is spacious and equipped with comfortable seating and workstations for students to relax and connect to the WIFI.
- **Video Conference Room:** Located centrally (6.5m x 11.50m) for virtual project meetings.

- **Hybrid Classroom:** Located on the bottom left (11m x 7.25m), optimized for best performance in teaching and learning.

- **Telecomm Closet (MDF):** Strategically placed near the central stairs and elevator to serve as the Main Distribution Facility (MDF).

- **Additional Facilities:** Includes a Surau (prayer room), gender-separated toilets, an elevator, and a wide main entrance foyer.

Second Floor (29.3m x 28m): Dedicated entirely to the core learning laboratories to minimize distractions, specialized teaching and learning and supporting the Faculty's goals for 4IR education. The design successfully places all four mandatory labs into the available footprint. All rooms adhere to the required dimensions(14m x 10m) and show arrangements for 30 workstations each. The Telecom Room on this floor is vertically aligned with the MDF on the 1st floor to ensure the most efficient vertical cable backbone.



- **4 New Labs:** Houses the 2 General Purpose Labs, 1 Cisco Network Lab, and 1 Embedded Lab. Each lab is placed in the corners of the expanded floor plan, meeting the strict 14m x 10m size requirement.

- **Capacity:** Each lab is fully equipped with the required 30 workstations and specific peripherals as outlined in the case study.

- **Central Hallway:** A spacious central hallway connects all labs, equipped with additional seating/study tables, toilets, and a secondary Telecomm Closet (IDF) aligned with the ground floor MDF for efficient vertical cabling.

REPORT APPENDIX

- **Appendix A: Projected Marks (Based on Task 1 Rubric)**

ITEM	DESCRIPTION	PROJECTED MARKS
Group name and members	Group name and members are clearly stated.	2 / 2
Suggested floor plan	Fits the description as per the case study provided.	2 / 2
Workstations & Equipment	A total of 30 workstations and other equipment planned for each lab.	2 / 2
Scale	Drawn with scale.	1 / 1
Clear and readable	Floor plan is clear, readable, and has appropriate labels.	2 / 2
Creativity	The layout design smartly separates communal areas (Floor 1) from dedicated learning labs (Floor 2) and includes practical additions like a Surau	1 / 1
TOTAL		**10 / 10**

- **Appendix B: Members' Task/Work Distribution**

  - **MUHAMMAD NABIL:** Analyzed the case study to ensure all room specifications (e.g., 14mx10m size and 30 workstations) were met. Drafted the main report document and compiled the report appendices

  - **MUHAMMAD AMIR:** Designed the building layout concept, drew the precise floor plan for Level 1 and Level 2 using Floorplanner, and determined the building scale.

  - **MOZAFER NASRELDEEN:** Acted as the secretary during group discussions to record meeting minutes and performed quality control (QC) against the grading rubric.

- **Appendix C: Meeting Minutes**

DATE/TIME	1 April 2026 / 17.00	LOCATION	Online via Discord/Google Meet
AGENDA	Floor plan design discussion and Task 1 delegation	Meeting MC	Muhammad Nabil GMeet

MEETING ATTENDANCE

NAME	ATTENDANCE	REASON FOR ABSENCE
Muhammad Nabil	Present	-
Muhammad Amir	Present	-
Mozafer Nasreldeen	Present	-

ITEMS DISCUSSED

NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Group Name Selection	Muhammad Nabil Rabbani suggested the name "The Chill Trio". Everyone agreed.	MUHAMMAD NABIL (30/3)
2	Floor Plan Concept	Muhammad Amir proposed a 21m x 21m layout. Verified that the Student Lounge is exactly 14m x 10m as required. Added a Surau and Toilets for practical use to gain creativity points.	MUHAMMAD AMIR (31/3)
3	Drawing Software	Agreed to use Floorplanner as the scale is easily adjustable.	MUHAMMAD AMIR (31/3)
4	Case Study Requirements	Mozafer Nasreldeen reminded the team that each lab must be 14mx10m and fit 30 workstations.	All Members (30/3)
5	PDF Report Compilation	The report will be compiled in Word and exported to PDF for submission before 5 PM on Thursday.	MUHAMMAD NABIL RABBANI (01/4)
6	Meeting Ended	22.30	